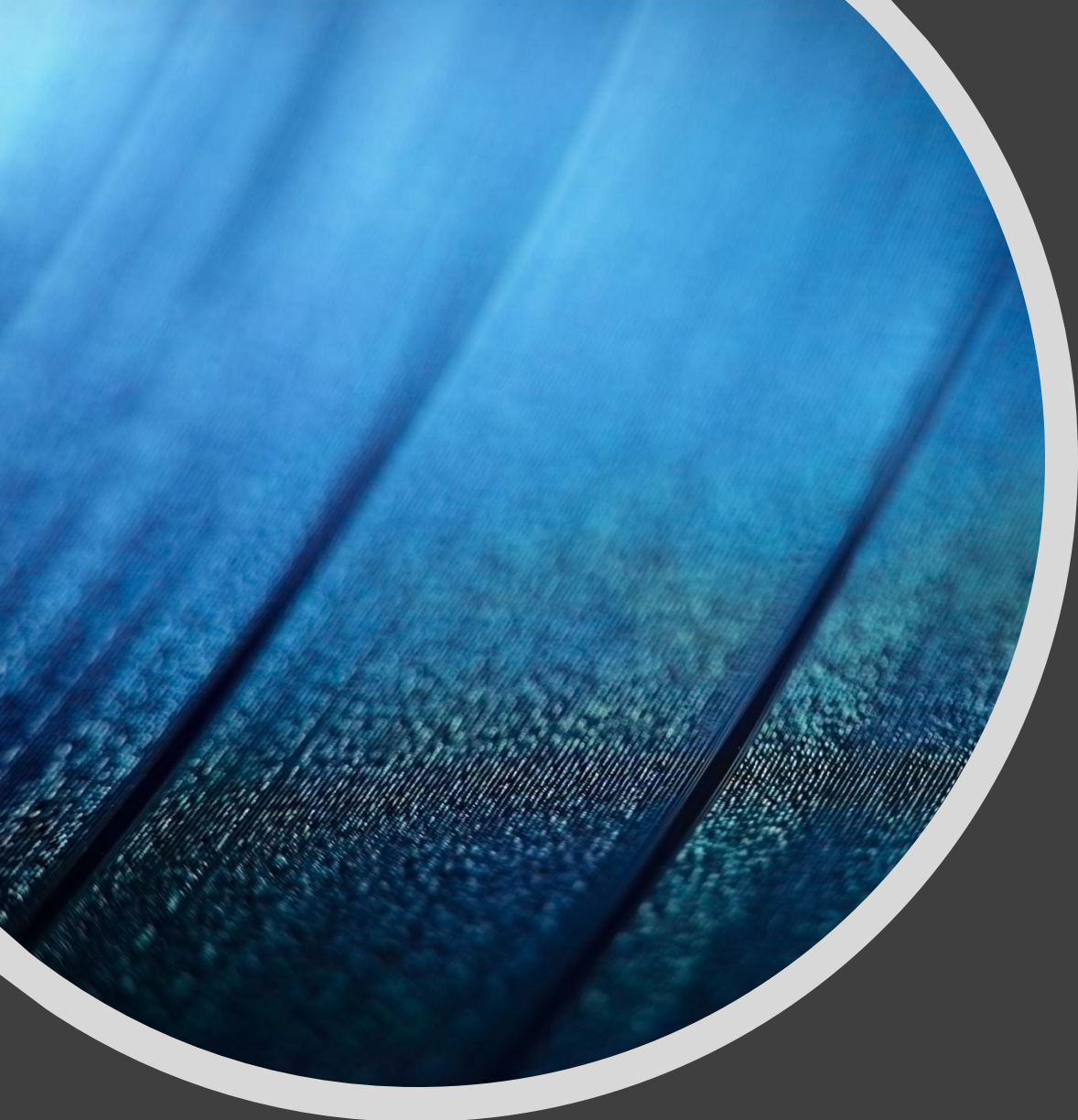


AI For the Better

Extra Power after releasing GPT & LLM technology



2024 // Warsaw IT-Days (Virtual) // Dennie Declercq



Content

Intro

Accessibility & Disabilities

General Purpose AI & A Better Life*

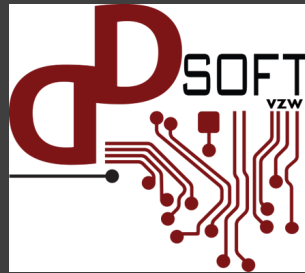
Let's HACK for The Better!

Outro



About Me

- Dennie Declercq
- President/ Developer DDSOFT
- Microsoft MVP
- @DennieDeclercq
- #DreamingIsBelieving
- <https://ware.ddsoft.be>



Accessibility & Disabilities

The Disability Spectrum

- Touch
 - One arm
 - Arm injury (T)
 - New parent (S)
- See
 - Blind
 - Low vision
 - Being distracted (T)
- Hear
 - Deaf
 - Hard of hearing
 - Ear infection (T)
- Speak
 - Non-verbal
 - Heavy accent
 - Mouth infection (T)
- Cognition
 - Learning disability
 - Down syndrome
 - Memory difficulties (S & T)
- Mental Health
 - Neurodiversity
 - Anxiety & Stress (S & T, or permanent)
 - Depression & Burnout (S or T)

T = Temporally
S = Situational

Learning Disability

Intellectual Functioning

Also called intelligence

Mental capacity

IQ age between 2 -8yo

Adaptive Behavior

Conceptual Skills

- Language/literacy
- Money
- Time/numbers

Social Skills

- Social responsibility
- Self-esteem

Practical skills

- Activities daily living
- Personal care / healthcare

Mental Health*

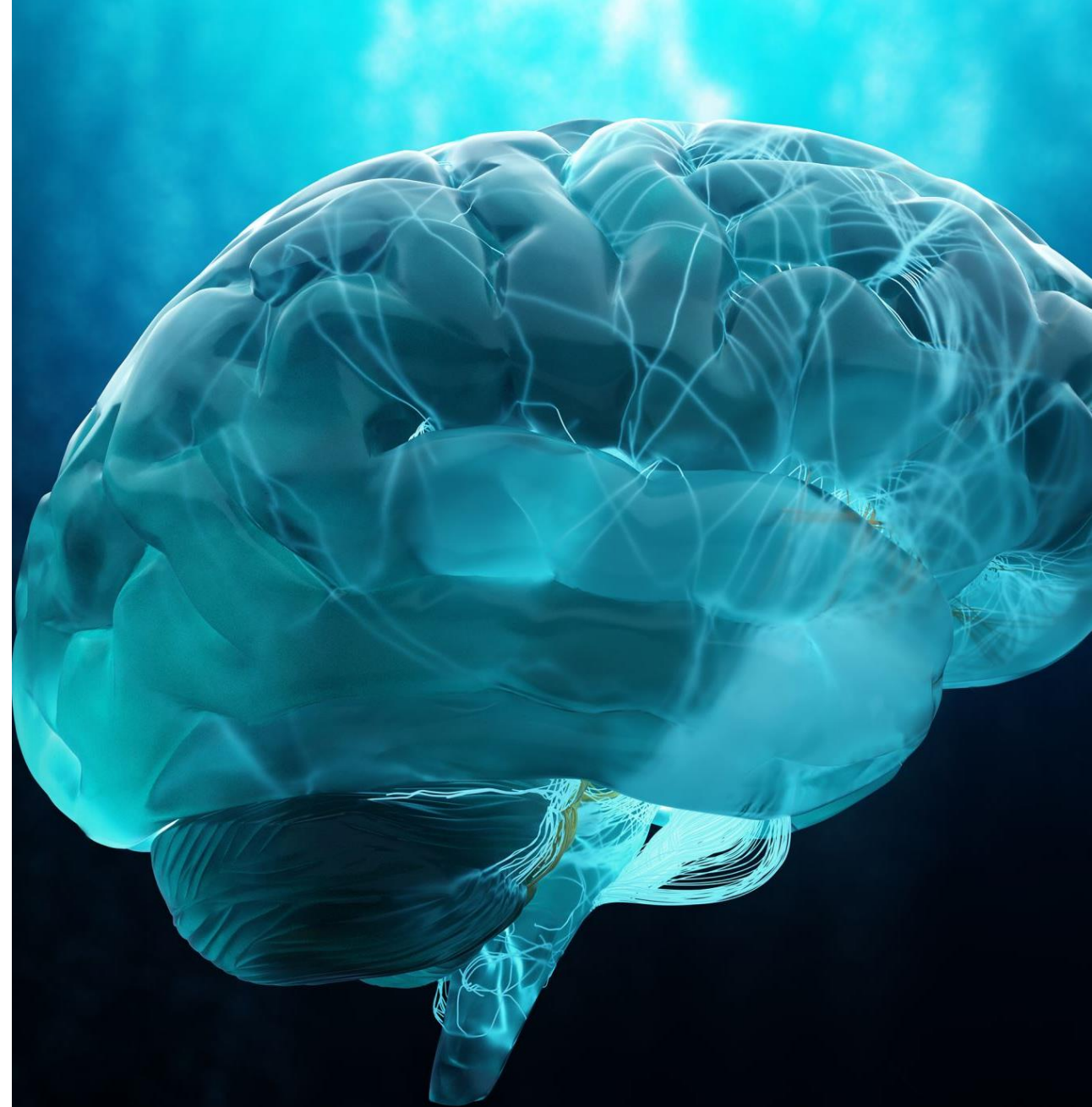
*= This list is not complete; Only meant as introduction in this topic.

- Stress, Anxiety, Depression, Burn-out
- Need for confirmations
- Need for motivation
- Exhaustion

Neurodiversity*

*= This list is not complete; Only meant as introduction in this topic.

- Autism
 - Structure
 - Confirmation
 - Slower stimuli-brain-response time
- Dyslexia
 - Difficulties writing, grammar & spelling
 - != less intellectual Capacity
- Dyscalculia
 - Difficulties with numbers
- AD(H)D
 - Motivation, concentration, structure and impulse-control



General
Purpose AI
&
“A Better Life*”

*= For People With Disabilities

OpenAI: Chat GPT

- Asking questions to a bot
- Give answers in 'human sentences'
- Uses context (few shot prompting)

! Cut-off the internet September 2021 !

Use cases FOR THE BETTER:

- ✓ Limit executive workload (Autism, ADHD,..)
- ✓ Write with good spelling (Dyslexia)
- ✓ Give Ideas

.....
.....
.....
.....



Bing COPILOT

- GPT style of working
- Recent data
- Ask questions and get reasoning
- Shows resources

Use cases FOR THE BETTER:

- ✓ Limit executive workload (Autism, ADHD,..)
- ✓ Write with good spelling (Dyslexia)
- ✓ Give Ideas



The screenshot displays the Bing COPILOT interface. At the top, a blue button contains the query "What is the population of Belgium?". Below this, two green checkmarks indicate the search process: "Zoeken naar 'population of Belgium'" and "Antwoorden voor u genereren...". The main response area shows the text: "As of September 3, 2023, the population of Belgium is 11,691,330 people ¹. The population density in Belgium is 386 per km² ¹." followed by "I hope that helps!". Below the response, a section titled "Meer informatie:" (More information:) shows "1 van 30" (1 of 30) results. Four sources are listed in blue boxes: "1. worldometers.info", "2. worldpopulationreview.com", "3. populationu.com", and "4. en.wikipedia.org". At the bottom, there are three additional query buttons: "What is the capital of Belgium?", "How does this compare to other European countries?", and "What are some famous landmarks in Belgium?".

GitHub CoPilot

- Code assistant
- Translate programming languages
- Write boiler code

Use cases FOR THE BETTER:

- ✓ Concepts with structure (Autism/ADHD)
- ✓ Reducing boring parts (Autism/ADHD)
- ✓ Help with new concepts (Autism/ADHD)

```
C# program.cs JS QuickSort.js •
JS > JS QuickSort.js > quickSort
1 // write a quicksort algorithm
2 // 1. pick a pivot
3 // 2. partition the array into two halves, one with
4 // 3. recursively call quicksort on the two halves
5 // 4. return the sorted array
6
7 // Write in JS (use recursion)
8
9 function quickSort(arr) {
10     if (arr.length <= 1) {
11         return arr;
12     }
13     let pivot = arr.shift();
14     let left = arr.filter(el => el < pivot);
15     let right = arr.filter(el => el >= pivot);
16     let leftSorted = quickSort(left);
17     let rightSorted = quickSort(right);
18     return [...leftSorted, pivot, ...rightSorted];
19 }
```

Microsoft Designer

- Make flyers and designs
- Make cards, leaflets,...

Use cases FOR THE BETTER:

- ✓ People with fantasy but without designer skills
- ✓ Entrepreneurs
- ✓ Solo developers
- ✓ (ex: on the Neuro Diversity Spectrum)



PROMPT: Make a flyer for a .NET community event about accessibility on September the 20th. The main speaker will be Dennie Declercq



Let's HACK for The Better

Deepfakes*

For The Better

- Deepfakes have a negative reputation
- Although the Deepfake technology can be used for THE BETTER
- Let's talk about Deepfakes in combination with
 - Rehabilitation
 - Motivation
 - Therapy
- *With or without VR



Deepfakes & Rehabilitation

- Seeing yourself moving around (F)
- Seeing yourself in a car (P)
- Seeing yourself performing a job (P / F)

F = Physical

P = Psychological



Deepfakes & Motivation

- Images of you in the house you want to live
- Images of you as a public speaker
- Newsflash of you reaching a goal



Deepfakes & Therapy

- Trauma Exposure Therapy
- Personal Growth
- Behavior training



CoPilots for The Better

- CoPilots are a productivity tool
- But the concept of Co-Pilots can also be used for personal and work life of:
 - People on the autism spectrum
 - People with Dyslexia
 - People with Learning Disabilities



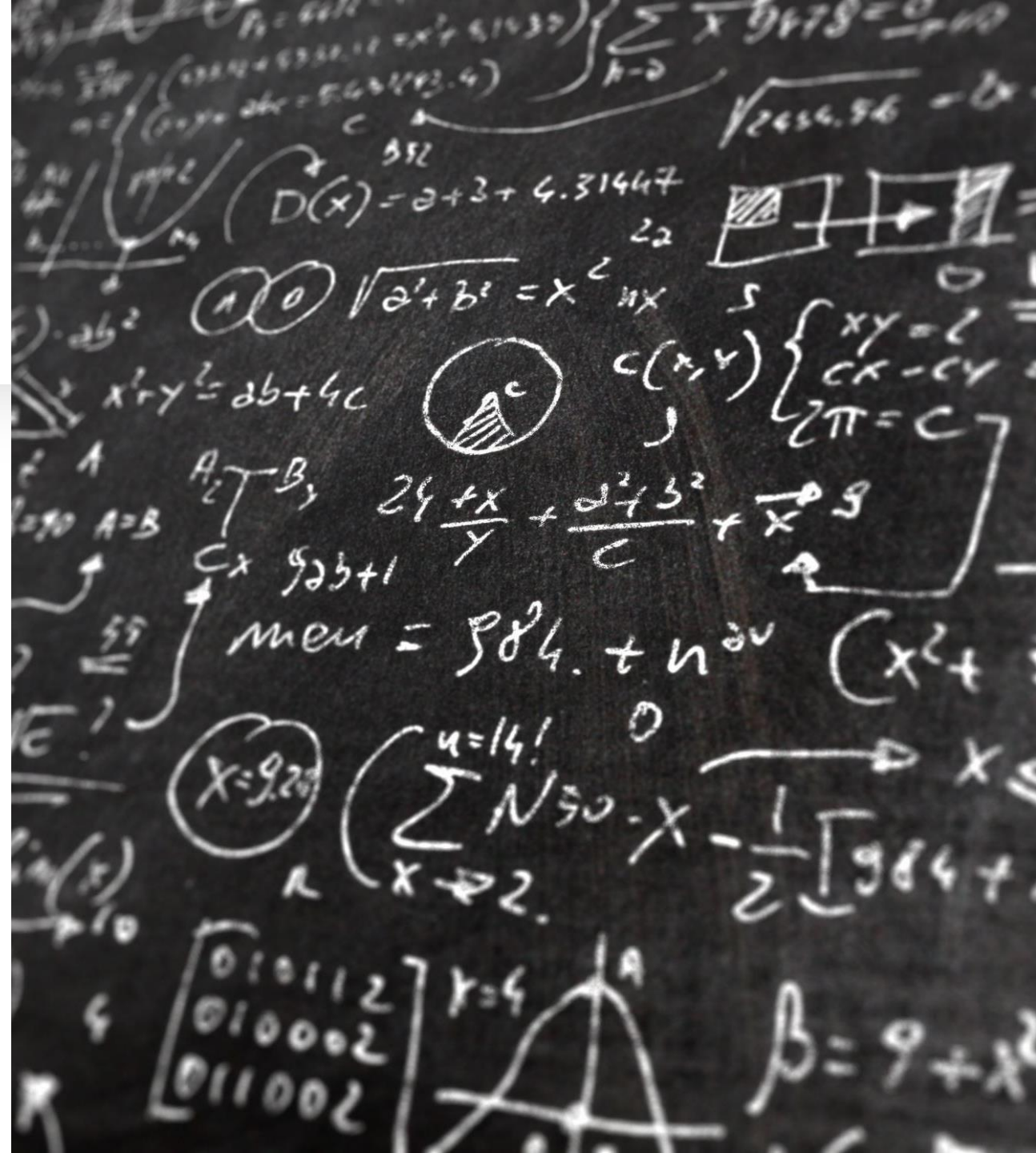
CoPilots & Autism

- Repetitive tasks
- Assistance with Planning & Administration
- Assistance with groceries shopping
- Assistance planning public transport



CoPilots & Dyslexia

- Helping with writing
- Assisting with summarizing
- Assisting with planning



CoPilots & Learning Disabilities

- Exercising money concept
- Assisting household- jobs
- Assist in transport
- Safeguard



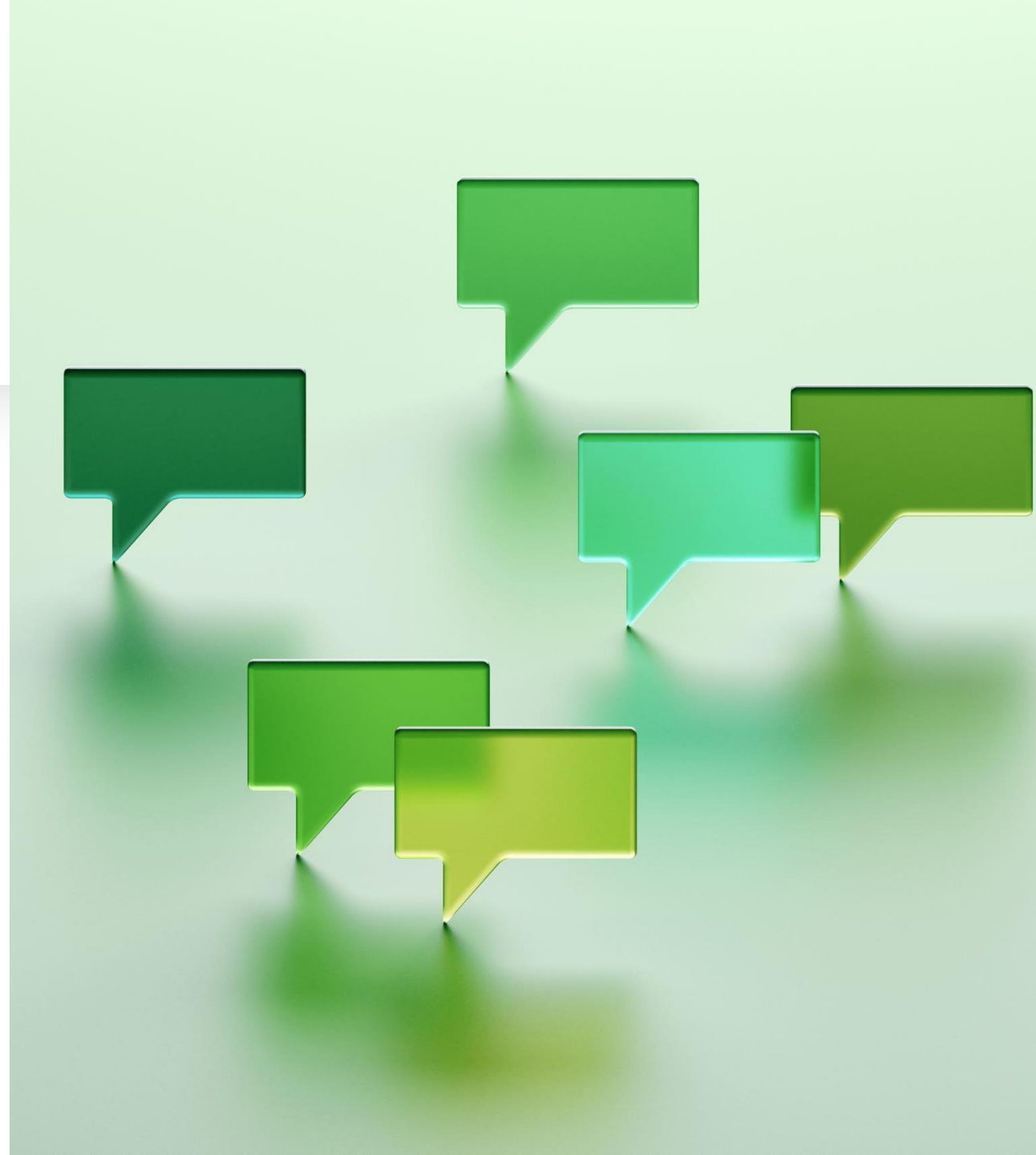
Chat Assistants for The Better

- Remember 2021 – 2022
 - I was ranting about Chatbots
- Look what a magic we have in our hands
- Let's use it FOR THE BETTER
- We are not there yet
- But we can start configure and finetune!



Chat Assistants & Autism

- Explaining scenario's multiple times
- Further reasoning than most folks do
- Confirming concepts multiple times



Chat Assistants & Mental Health

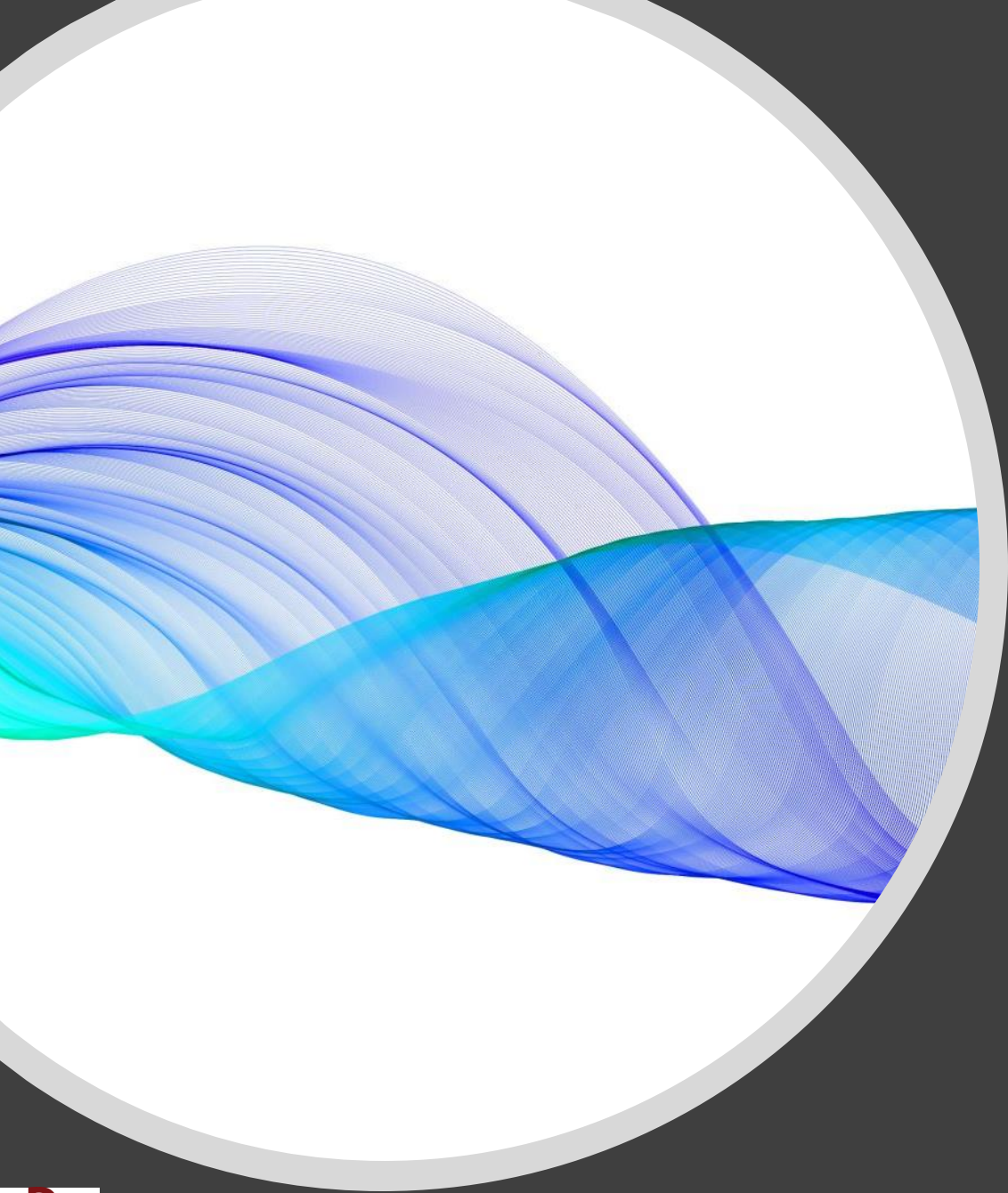
- Giving confirmations
- Someone with interest
- Therapy for those who cannot go to a psychotherapist*

*= Yes.. I know this is taboo!



Chat Assistants & Learning Disabilities

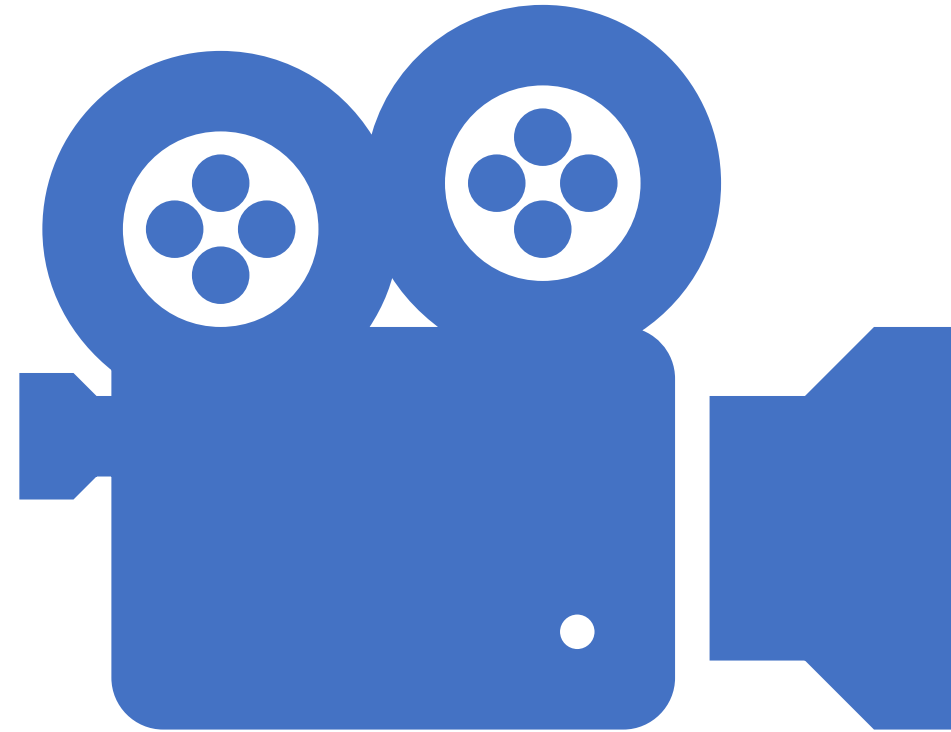
- Explaining societal concepts in other jargon
 - "Explain to a kid"
- Rehearse words & Jargon
- Train historic information



Recap

Resources

- Recording AI for The Better (original):
<https://go.ddsoft.be/ai4thebetter>
- Slides AI for the Better (Original):
<https://go.ddsoft.be/AI4TheBetterSlides>
- Slides AI for the Better: Extra Power after releasing GPT & LLM Technologies
<https://go.ddsoft.be/ai4thebetterGPT>
- <https://ware.ddsoft.be>





Thanks || Q&A

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- dennie@ddsoft.be
- <https://ware.ddsoft.be>



FEEDBACK

AI For The Better: Extra Power after releasing GPT



Dennie Declercq

<https://warszawskiedniinformatyki.pl/user.html#!/lecture/WDI24-a132/rate>

@DennieDeclercq - <https://ware.ddsoft.be>